

```

/*****
* AUTHOR:      CYRIL ANDRE B. DURANGO
* CLASS:       CPE 3101L - INTRODUCTION TO HDL
* SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
* -----
* PROJECT FILE TITLE: Problem_A.v
*
*****/

module Problem_A
(
    input wire  En, nReset, clk,
    output reg  [5:0] Q
);
    reg counter;

    always @(negedge clk, negedge nReset)
    begin
        if (!nReset)
            Q <= 6'd0;
        else if (En) begin
            Q <= {~Q[0], Q[5:1]};
        end
        else if (!En)
            Q <= Q;

    end

endmodule

```

Quartus Prime Lite Edition - Z:/Finals/Problem_A/Problem_A - Problem_A

File Edit View Project Assignments Processing Tools Window Help

Problem_A_v tb_Problem_A_v Compilation Report - Problem_A tb_BinaryCounter_4Bit.v

Entity/Instance: MAX 10-10M08DAF484C8G

Problem_A

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
- EDA Netlist Writer
- Flow Messages
- Flow Suppressed Messages

Flow Summary

Flow Status: Successful - Fri Dec 13 08:40:47 2024

Quartus Prime Version: 20.1.1 Build 720 11/11/2020 SJ Lite Edition

Revision Name: Problem_A

Top-level Entity Name: Problem_A

Family: MAX 10

Device: 10M08DAF484C8G

Timing Models: Final

Total logic elements: 7 / 6,064 (<1%)

Total registers: 6

Total pins: 9 / 250 (4%)

Total virtual pins: 0

Total memory bits: 0 / 387,072 (0%)

Embedded Multiplier 9-bit elements: 0 / 48 (0%)

Total PLLs: 0 / 2 (0%)

UFM blocks: 0 / 1 (0%)

ADC blocks: 0 / 1 (0%)

Tasks

Compilation

Task

- Compile Design
- Analysis & Synthesis
- Filter (Place & Route)
- Assembler (Generate program)
- Timing Analysis
- EDA Netlist Writer
- Edit Settings
- Program Device (Open Programmer)

Messages

System (23) Processing (121)

100% 00:00:23

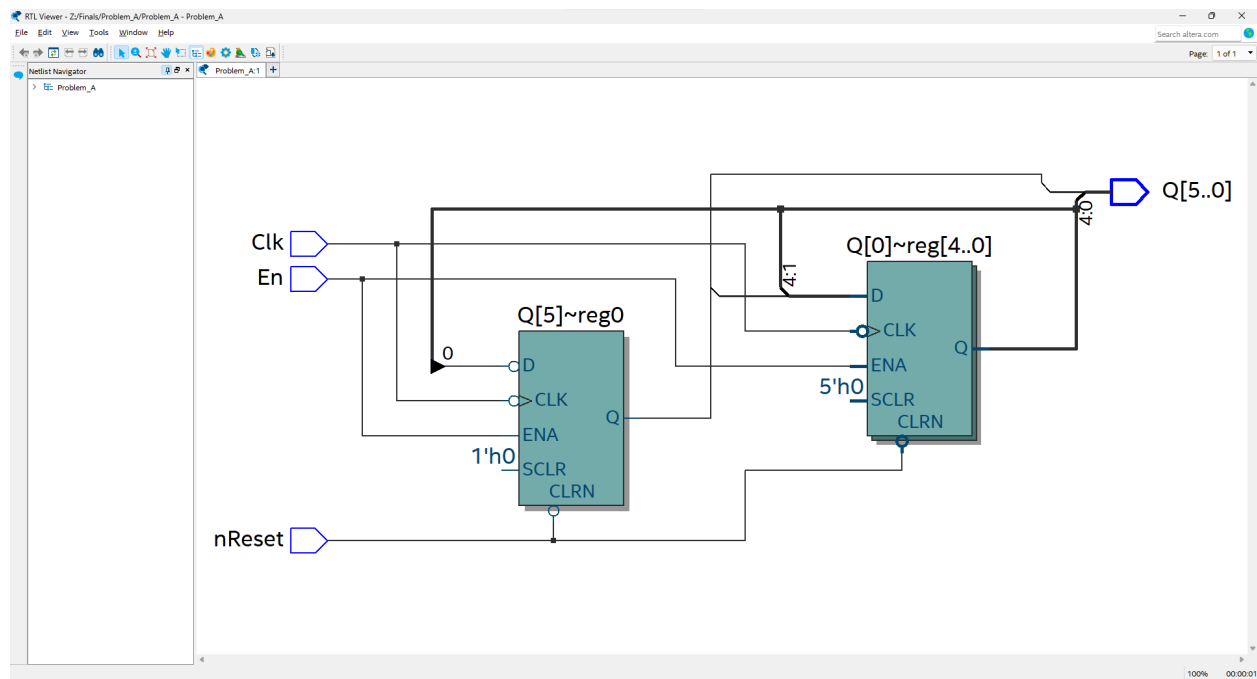
Command: quartus_eda --read_settings_files=off --write_settings_files=off Problem_A -c Problem_A

18236 number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NPL_PARALLEL_PROCESSORS in your QSF to an appropriate value for best performance.

204019 generated file Problem_A.vo in folder "Z:/Finals/Problem_A/stimulation/modelsim/" for EDA simulation tool

Quartus Prime EDA Netlist Writer was successful. 0 errors, 1 warning

293000 Quartus Prime Full compilation was successful. 0 errors, 13 warnings



```

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5  * -----
6  * PROJECT FILE TITLE: tb_Problem_A.v
7  *
8  *****/
9
10 `timescale 1 ns/1 ps
11 module tb_Problem_A();
12     reg      En, nReset, clk;
13     wire [5:0] Q;
14
15     Problem_A UUT(En, nReset, clk, Q);
16
17     initial
18         clk = 1'b0;
19
20     always
21         #2 clk = ~ clk;
22
23     initial begin
24         nReset = 1'b0; #16
25         nReset = 1'b1;
26     end
27
28
29     initial begin
30         En = 1'b0; #8
31         En = 1'b1; #8
32         En = 1'b0; #8
33         En = 1'b1; #24
34         En = 1'b0; #8
35         En = 1'b1; #24
36     end
37
38
39     $stop;
40 end
41
42
43
44 endmodule

```

